Drawing Conclusions from Measurement

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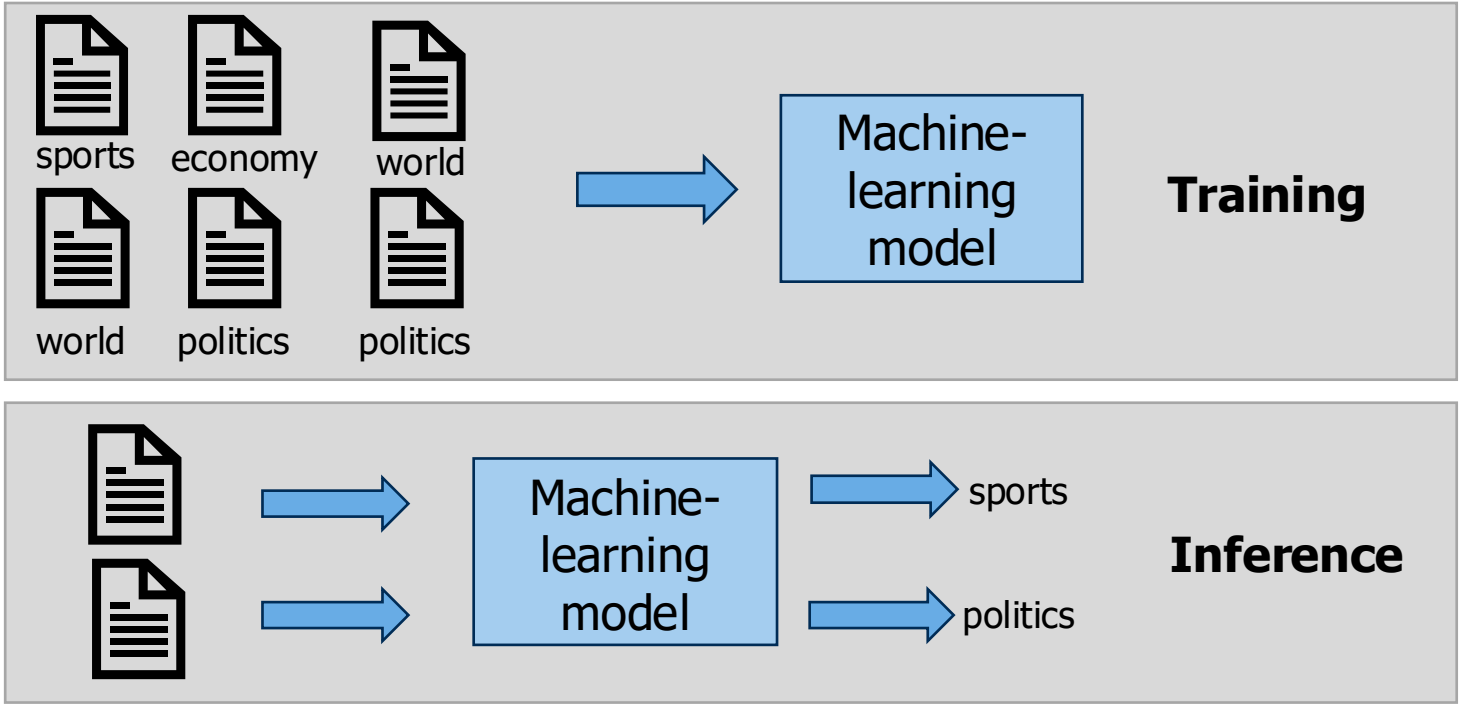
Methods of Data analysis

- We want to know if (and when and how) Republicans talk about taxes more than Democrats:
  1. We use word statistics to find if words like "taxes" and "spending" are more common in republican speeches
  2. We can train a topic model, identify the tax-related topics and determine if that topic is more common in Republican vs. Democratic speech (or incorporate party affiliation as co-variate in STM)
  3. **We could go through every speech by hand:**
     - **Label if each speech or sentence or word is related to taxes**
     - **Count if we labeled more Republican speech than Democratic speech**
  4. **We can automate #3 using machine learning**

Supervised learning



Supervised learning

- We represent each document as features:
  - E.g. TF-IDF weighted BOW features
  - Frequencies of lexicon words
  - Word embeddings
- We train a supervised classifier to predict document labels:
  - Logistic regression classifier
  - Neural model (with initial embedding layer): RNN, LSTM, CNN, Transformer
  - Review of supervised learning:
    - Jurafsky & Martin Chap. 5
    - Jurafsky & Martin Chap. 7
- [alternative: we use an LLM to predict document labels]

Drawing Conclusions from Measurements

- Running example: do Republicans talk more about taxes than Democrats?
  - With document labels, we can estimate the difference, but how can we tell if the estimated difference is meaningful?

- [An aside: what if our model predictions are not very good? We'll return to this question in a few weeks]

Overview

- Hypothesis Testing:
  - Background
  - Examples
- Additional Metrics:
  - Effect size
  - Confidence intervals
- Starting next topic: Causal Inference

Hypothesis Testing

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Formalizing Hypotheses

- Formalize a research question into testable hypothesis

Question	Hypothesis
What strategies does the Russian government use to manipulate public opinion?	State-affiliated and independent news outlets in Russia covered the Russia-Ukrainian War at different rates

Formalizing Hypotheses

- Null hypothesis: a claim, assumed to be true, that we'd like to test (because we think it is wrong)

Question	Hypothesis	Null Hypothesis
What strategies does the Russian government use to manipulate public opinion?	State-affiliated and independent news outlets in Russia covered the Russia-Ukrainian War at different rates	Level of coverage is the **same**

Key Idea

- If the null hypothesis were true, how likely is it that you'd see this data?

Example:

- Hypothesis: Baltimore residents tend to be politically liberal
- H_0: Among all N registered {Democrat, Republican} primary voters, there are an equal number of Democrats and Republicans in Baltimore

$$\frac{\#dem}{N} = \frac{\#rep}{N} = 0.5$$

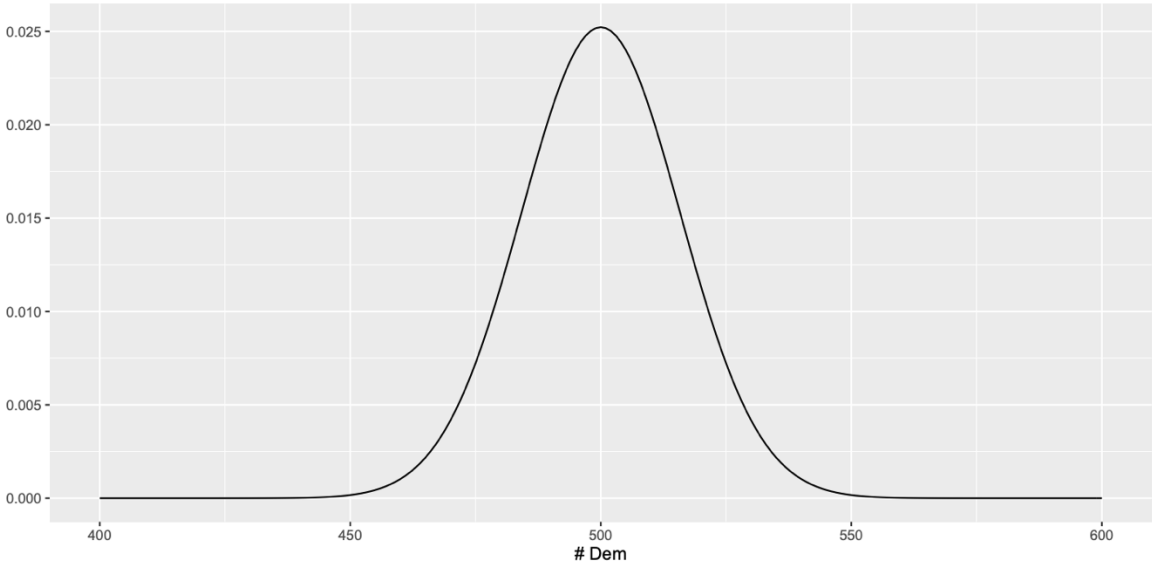
Hypothesis Testing

- If we had access to the party registrations (and knew the population), we would have our answer.
- Hypothesis testing measures our confidence in what we can say about a null *from a sample*

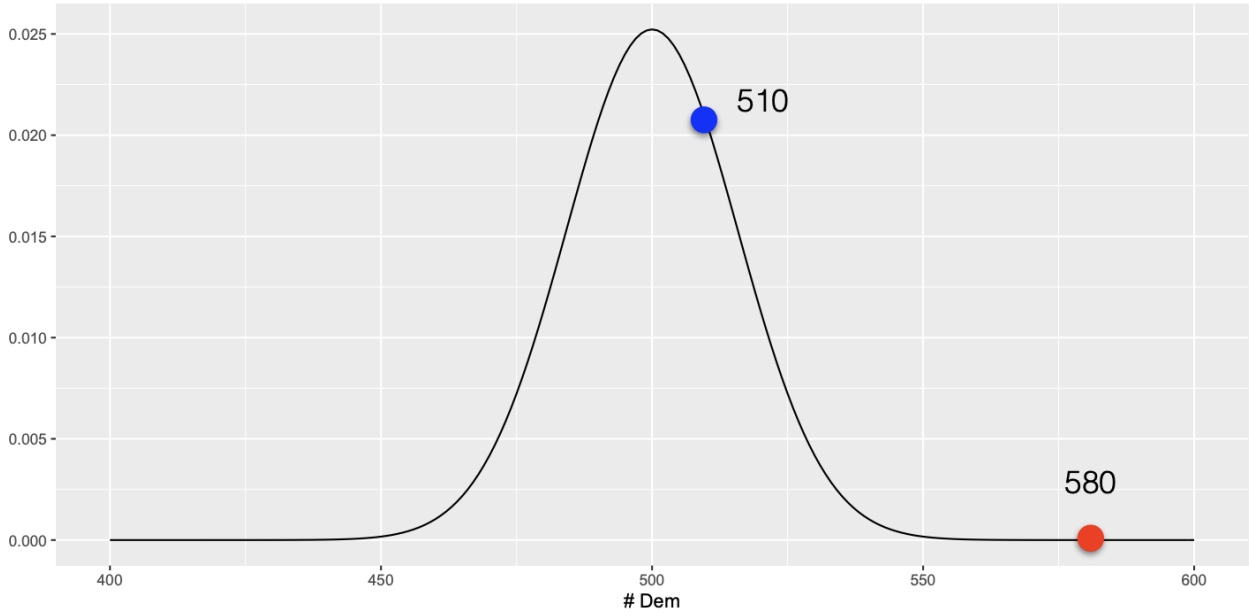
- Let's say we have enough research money to survey 1000 people. We get 510 of respondents saying they are Democrats. Can we conclude that Baltimore is left-leaning?

Hypothesis Testing: Example

- If the null hypothesis is true, Baltimore is 50% Democratic
- Surveying 1000 people, is the same as drawing from a binomial probability distribution with n=1000 with p = 0.5



Hypothesis Testing: Example



At what point is a sample statistic unusual enough to reject the null hypothesis?

Hypothesis Testing: Example

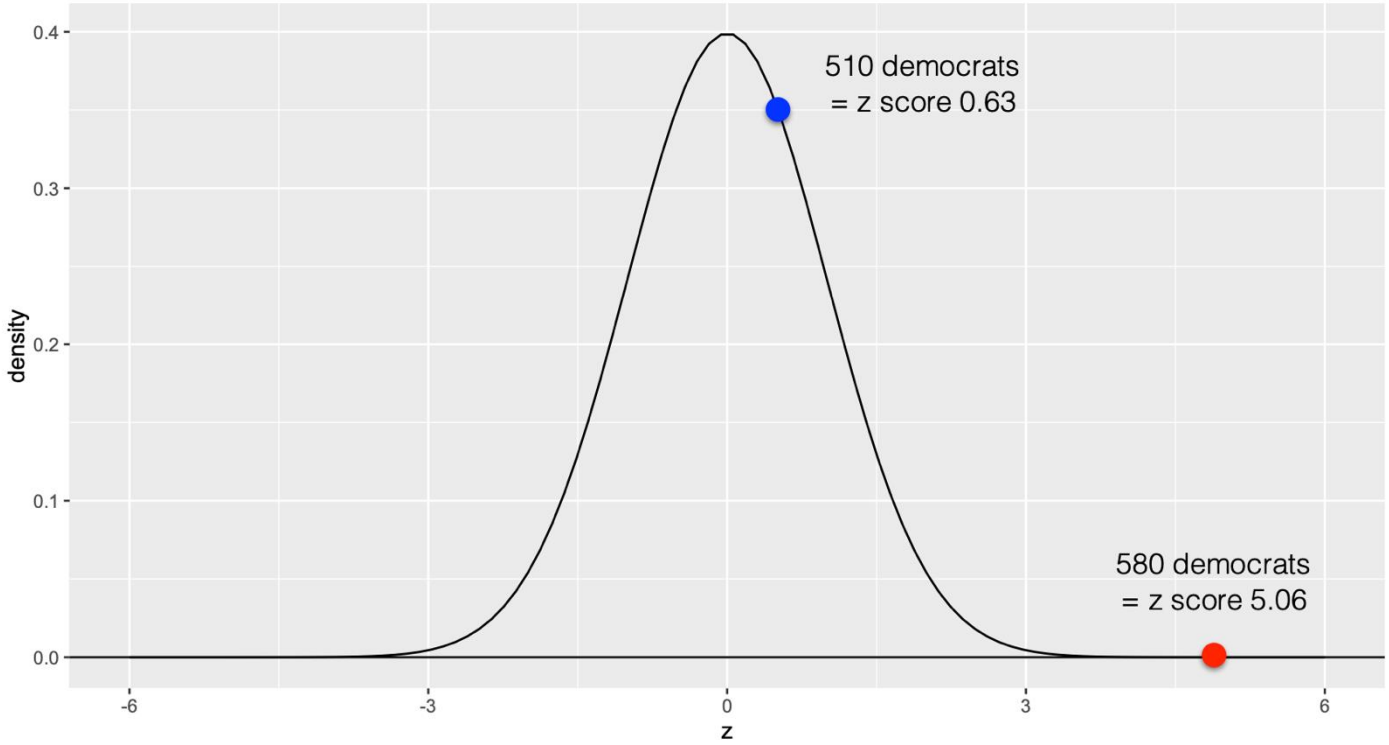
- The form we assume for the null hypothesis lets us quantify that level of surprise.
- We can do this for many parametric forms that allows us to measure $P(X \leq x)$ for some sample of size
- For large n, we can often make a normal approximation.

Compute a test statistic

- Test statistic: a function of the sample data that we will use to decide if we should reject the null
- Common choice: Z-score
  - Number of standard deviations away a data point or observed value is from the mean
  - $Z = \frac{X-\mu}{\sigma}$
  - [for Normal distributions, transform into standard normal (mean = 0, standard deviation =1)]
- For Binomial distributions, normal approximation (for large n)
  - $Z = \frac{Y-np}{\sqrt{(np(1-p)}}$

    Y = 580 (democrats in sample)
    p = 0.5 (proportion we are testing)
    n = 1000 (sample size)

Hypothesis Testing: Example



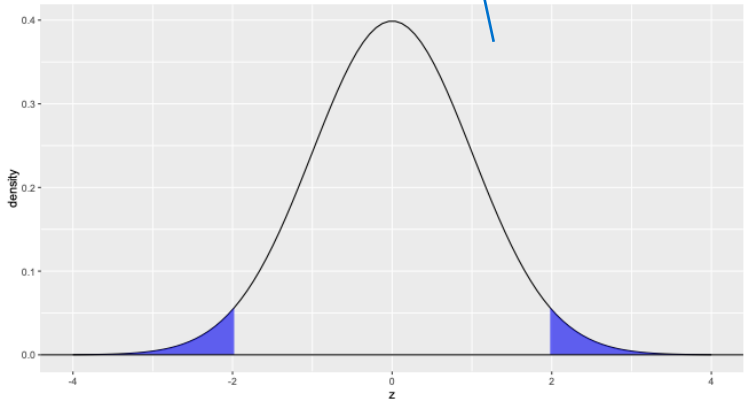
Significance Testing

- Decide on the level of significance α {0.05, 0.01}
  - 0.05: we reject the null when the probability of getting a sample mean is less than 5% if the null were true
- Testing is evaluating whether the sample statistic falls in the rejection region defined by α

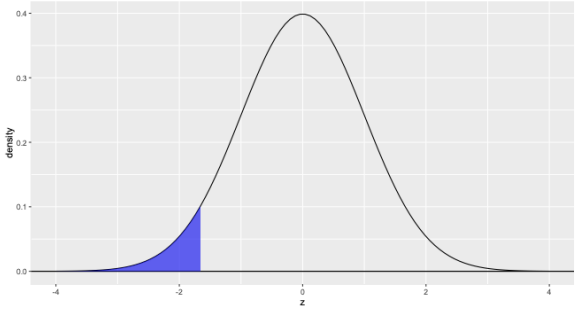
Tails

- Two-tailed tests measured whether the observed statistic is different (in either direction)
- One-tailed tests measure difference in a specific direction
- All differ in where the rejection region is located; α = 0.05 for all.

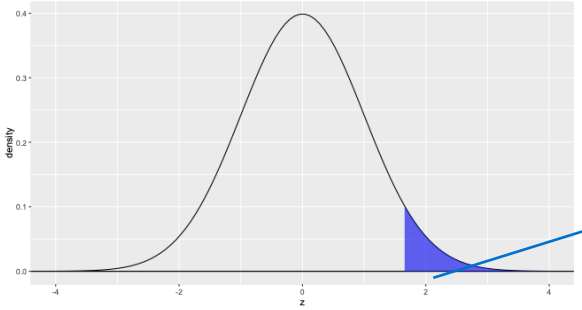
Baltimore tends to be politically biased



two-tailed test



lower-tailed test



upper-tailed test

Baltimore tends to be liberal

P-value

- A p-value is the probability of observing a statistic at least as extreme as the one we did if the null hypothesis were true.
- Lower-tailed test
  - p-value(z) = $\mathrm{P}(Z \leq z)$
- Two-tailed test
  - p-value(z) = $2 * \mathrm{P}(Z \leq -|z|)$
- Upper-tailed test
  - p-value(z) = $1 - \mathrm{P}(Z \leq z)$

z: observed test statistic
Z: test statistic

Hypothesis testing process

1. Formulate the null hypothesis (H_0) and alternative hypothesis (H_1)
2. Choose a Significance Level (α), the threshold for rejecting H_0 (e.g., 0.05)
3. Select an appropriate statistical test based on the data and study design
4. Compute the test statistic
5. Find the p-value
6. Reject H_0 if the p-value $\leq \alpha$ otherwise, fail to reject H_0

Choosing a Parametric hypothesis test

Assumes data is normally distributed

	Predictor variable	Outcome variable	Research question example
Paired t-test	• Categorical • 1 predictor	• Quantitative • groups come from the same population	What is the effect of two different test prep programs on the average exam scores for students from the same class?
Independent t-test	• Categorical • 1 predictor	• Quantitative • groups come from different populations	What is the difference in average exam scores for students from two different schools?
ANOVA	• Categorical • 1 or more predictor	• Quantitative • 1 outcome	What is the difference in average pain levels among post-surgical patients given three different painkillers?
MANOVA	• Categorical • 1 or more predictor	• Quantitative • 2 or more outcome	What is the effect of flower species on petal length, petal width, and stem length?

Z-test if sample size is large or variance is known

https://www.scribbr.com/statistics/statistical-tests/

Errors

		Test Results	
		Keep null	Reject null
Truth	Keep null		Type I Error
	Reject null	Type II Error	

- For any significance level α and n hypothesis tests, we can expect α×n type I errors.
- α=0.01, n=1000 = 10 "significant" results simply by chance

Multiple Hypothesis Correction

- Bonferroni correction:
  - For family-wise significance level α_0 with n hypothesis tests:

$$a \leftarrow \frac{a_0}{n}$$

  - Very strict; controls the probability of at least one type I error.
  - Less strict alternative: Holm method

- Alterative: Benjamini-Hochberg
  - Less strict, controls the expected proportion of false discoveries
  - [accepts that some discoveries may be false but limits their proportion]

Some examples

Example statistical tests in the wild

- Research question:
  - Do Wikipedia articles perpetuate gender and racial bias and stereotypes?

- Hypothesis:
  - Wikipedia articles about women have longer “personal life” sections
  - Wikipedia articles about men have longer “career” sections
  - [Null: section sizes are the same]

Anjalie Field, Chan Young Park, Kevin Z. Lin, and Yulia Tsvetkov. "Controlled Analyses of Social Biases in Wikipedia Bios" WebConf (2022)

Why do we care about Wikipedia?

- Extremely widely read platform-–bias and stereotypes on Wikipedia can influence readers

- Wikipedia is widely used as training data in NLP models
  - We know that models are liable to absorbing and amplifying data biases

- In some aspects, Wikipedia can be seen as a reflection of society – we can learn about bias and stereotypes in society by studying Wikipedia

- Bonus: Wikipedia has an active editor community, and studies that have revealed biases on the platform have motivated the editor community to correct them [Langrock and Sandra González-Bailón 2020; Reagle and Rhue 2011]

Example statistical tests in the wild

- Method:
  - Pair each article about a women with an article about a man with similar attributes (e.g. profession, nationality)
  - For each article, compute the % of words in each subsection
    - Controls for variable article length
  - Compare proportions of subsections

- Significant test:
  - Paired t-test
  - Multiple hypothesis correction

Results for “personal”, “life”, “early” and “career” sections

Section	Average % women	Average % men	p-value
career	**17.47%**	15.55%	4.20E-41
early life and education	**1.43%**	1.09%	1.08E-31
personal life	**2.47%**	2.12%	1.35E-25

Women have longer “career” and “life” sections?

Example statistical tests in the wild

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Example statistical tests in the wild

- Research question:
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- Hypothesis:
  - Wikipedia articles about women have longer “personal life” sections
  - Wikipedia articles about men have longer “career” sections
  - [Null: section sizes are the same]

- Conclusion:
  - We do find significance differences in article sections
  - Men tend to have fine—grained career descriptions, while women have more generic career descriptions and more words discussing “personal life”

Example statistical tests in the wild

- Research question:
  - Do Wikipedia articles reflect language and nationality bias?
  - "Local heroes" effect

- Hypothesis:
  - Articles about Hispanic/LatinX Americans are available in fewer languages than articles about non-Hispanic/LatinX people
  - Articles about Hispanic/LatinX Americans are available in Spanish more often

- Which statistical test to use?
  - t-test/z-test are for continuous data where we can assume a normal distribution
  - Test for *paired categorial* data: McNemar's Test

Example statistical tests in the wild

	Hispanic/LatinX	**Non-Hispanic/LatinX**	P-value
# of Languages	7.54	7.62	0.668
% available in Spanish	34.09%	27.54%	2.15E-12

Paired t-test, not significant

McNemar's test, significant

Non-parametric hypothesis tests

	Predictor variable	Outcome variable	Use in place of...
Spearman's *r*	• Quantitative	• Quantitative	Pearson's *r*
Chi square test of independence	• Categorical	• Categorical	Pearson's *r*
Sign test	• Categorical	• Quantitative	One-sample *t*-test
Kruskal–Wallis *H*	• Categorical • 3 or more groups	• Quantitative	ANOVA
ANOSIM	• Categorical • 3 or more groups	• Quantitative • 2 or more outcome variables	MANOVA
Wilcoxon Rank-Sum test	• Categorical • 2 groups	• Quantitative • groups come from different populations	Independent t-test
Wilcoxon Signed-rank test	• Categorical • 2 groups	• Quantitative • groups come from the same population	Paired t-test

https://www.scribbr.com/statistics/statistical-tests/

Considerations for choosing a hypothesis test

- If data is continuous, categorical, or binary
- Sample size
- If data can be assumed to be normally distributed (can depend on sample size)
- Number of variables to be analyzed
- If data is paired

Effect size and confidence intervals

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Other useful ways of conveying results

- Effect size
- Confidence intervals
  - Bootstrap

Effect Size

- Hypothesis testing tells us if difference exists – not how big the differences is

- Common measurement: Cohen's d:
  - difference in means divided by (pooled) standard deviation of data
  - $\frac{avg(x_1) - avg(x_2)}{s}$

d	size
0.01	Very small
0.2	small
0.5	medium
0.8	large
1.2	Very large
2	huge

Sawilowsky, S (2009). "New effect size rules of thumb". *Journal of Modern Applied Statistical Methods*. **8** (2): 467–474. doi:10.22237/jmasm/1257035100

Confidence Intervals

- Interpretation:
  - The probability that a parameter will fall within a particular range X% of the time
  - 95% confidence interval: If we repeat this experiment infinite times, 95% of the time, the outcome will be in range we computed
  - Handwave: we're 95% certain this range contains the true value [it could be the 5% of experiments where we got the wrong range]
  - The 95% confidence interval for an effect will exclude the null value if and only if the test of significance yields a p value of less than 0.05

Confidence Intervals: Computation

$$CI = \bar{X} \pm z\frac{\sigma}{\sqrt{n}}$$

$\bar{X}$: sample mean

z : controls size of confidence interval

σ : population standard deviation

n : sample size

Requires that sample is random, variable of interest has a normal distribution, and that we know population standard deviation

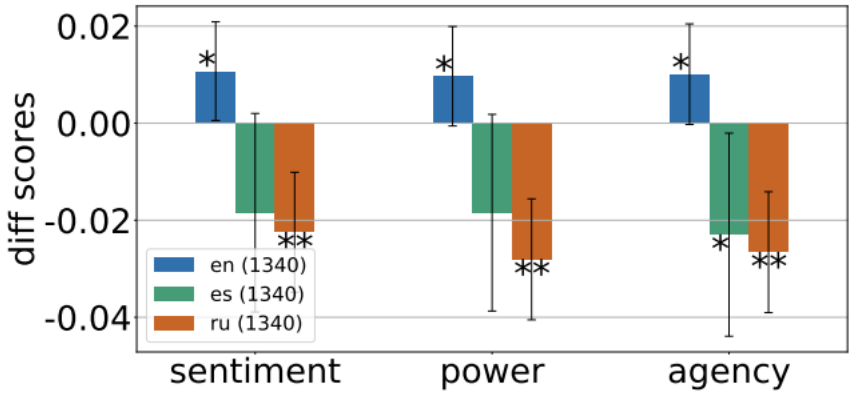
Bootstrap

- Resampling process for estimating summary statistics

- Choose number of bootstrap samples and sample size
  - For each bootstrap sample:
  - Draw a sample with replacement with the chosen size
  - Calculate the statistic on the sample
- Calculate the mean of the sample statistics. Can calculate confidence intervals from ranked statistics

- Can be used to evaluate predictive power of ML models: bootstrap sampling over the *training* set and retrain model over each sample

Example: Connotations about LGBT people in Wikipedia articles

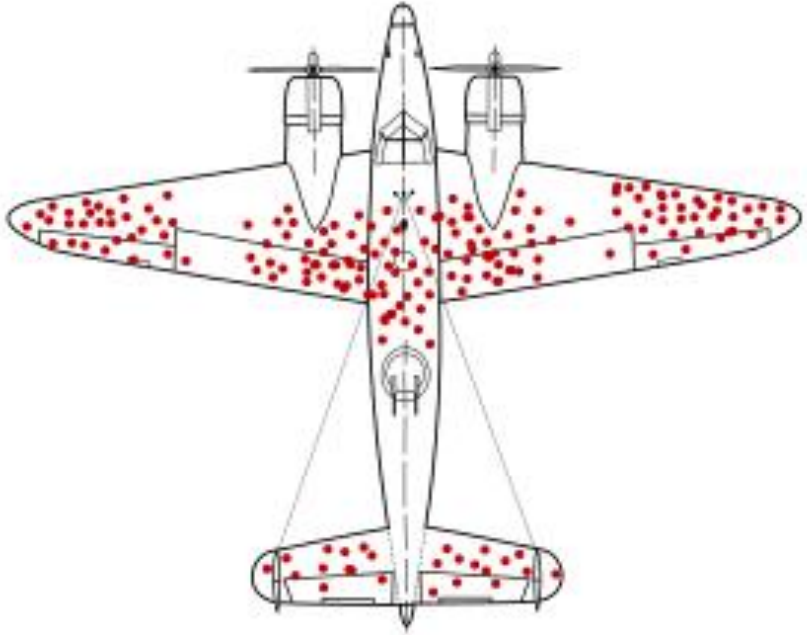


- English articles: LGBT people are portrayed with more **positive** connotations than non-LGBT people
- Spanish articles: results are inconclusive with wide confidence intervals
- Russian articles: LGBT people are portrayed with more **negative** connotations than non-LGBT people
- [Significance measured using paired t-test]

Hypothesis testing can't guarantee your results are valid!

- Examples:
  - Selection/sampling/survivorship bias
  - Simpson's paradox

Survivorship Bias



Example from Wikipedia

- Numerous studies have shown that Wikipedia articles about women tend to be longer than articles about men [Graells-Garrido et al. 2015; Reagle & Rhue 2011; Wagner et al. 2015; Young et al. 2016]

- Naïve conclusion:
  - Wikipedia editors are biased towards women and spend more time writing detailed accounts of their lives

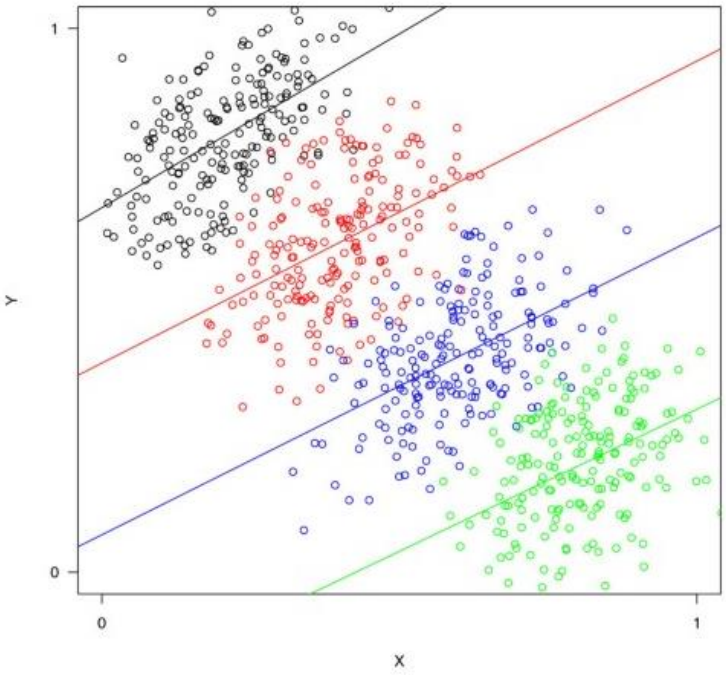
- Hypothesis [Wagner et al. 2015]
  - There is a higher bar for women to have Wikipedia articles – only the most famous women with lots of accomplishments have Wikipedia articles
  - "Glass ceiling effect"

Simpson's Paradox

	Total Average Length
Articles about men	587.85 (n = 342,677)
Articles about women	634.89 (n = 108,915)

Articles about women are longer (significantly)

Simpson's Paradox



- Data looks negatively correlated overall
- Subsetting data shows positive correlations

https://bmcmedresmethodol.biomedcentral.com/articles/10.1186/1471-2288-8-34

Simpson's Paradox

	Total Average Length	Articles about non-athletes	Articles about athletes
Articles about men	587.85 (n = 342,677)	900 (n = 142,677)	365.17 (n = 200,000)
Articles about women	634.89 (n = 108,915)	600 (n = 50,000)	195.74 (n = 58,915)

- It's true that Wikipedia articles about men are dominated by athlete "stubs"
- It's true that when you control for things like "is the article a stub" and "is the person an athlete", you no longer see articles about women are longer
- [I made up the actual numbers]

Takeaways

- Basic idea behind hypothesis testing
- Ability to interpret use of hypothesis tests by others
- Ability to select a hypothesis test based on data characteristics
- Selection bias, Simpson's paradox

- Where to learn more?
  - Statistics classes

- Next class: causal inference
  - How can we make sure we are measuring the right thing?

Causal Inference: Definitions

What is causal inference?

- Process of (1) establishing, and (2) quantifying causal relationships empirically (using statistics + data)

- Classic setup: is cold medicine effective?

Treatment T
t = took cold medicine
t = did not take medicine

Outcome Y
y = had fever
y = did not have fever

Some notation: Potential Outcomes

- T: Treatment
- Y: Outcome

- Y(t): The *potential outcome* you would observe under treatment t
  - Outcome that you can *potentially* observe, but that you may not actually observe

- Estimands: causal effects that we want to measure from the data

Causal Estimand: Individual Treatment Effect (ITE)

- For each individual *i*,
  - ITE = $Y_i(t = 1) - Y_i(t = 0)$
  - [Outcome had person *i* taken cold medicine – outcome if they did not]

- ITE is often what we actually care about: should you take cold medicine?

- But we can't measure it!
  - Either you take the medicine or you don't: we can't observe both $Y_i(t = 1)$ and $Y_i(t = 0)$
  - **Fundamental problem of causal inference**
  - Once an outcome is observed, the unobserved outcome is called the *counterfactual*

Causal Estimand: Average Treatment Effect (ATE)

i	T	Y	$Y(1)$	$Y(0)$	$Y(1) - Y(0)$
1	0	0	?	0	?
2	1	1	1	?	?
3	1	0	0	?	?
4	0	0	?	0	?
5	0	1	?	1	?
6	1	1	1	?	?

- ATE = $E[Y_i(T = 1) - Y_i(T = 0)] = E[Y(1)] - E[Y(0)]$
- Does ATE = $E[Y \mid T = 1] - E[Y \mid T = 0]$?
  - Can we just average over the data in the table, ignoring the missing values?

Image credit: https://www.bradyneal.com/Introduction_to_Causal_Inference-Dec17_2020-Neal.pdf

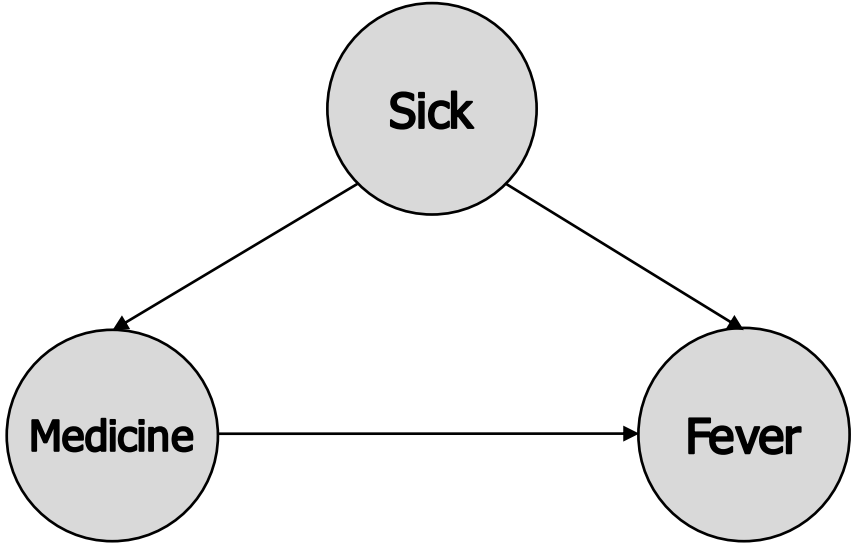
Does ATE = $E[Y \mid T = 1] - E[Y \mid T = 0]$?

- Let's pretend we surveyed a bunch of people. We asked them if they took medicine on Sunday and if they had a fever on Monday

		Has Fever (Y)		
		Yes	**No**	**Sum**
Took Medicine (T)	**Yes**	61	43	104
	No	12	80	92
	Sum	73	113	186

- $E[Y = fever \mid T = medicine] - E[Y = fever \mid T = no\ medicine]$
- $\frac{61}{104} - \frac{12}{92} = 0.4561$
- Value is positive → taking medicine causes fevers?!

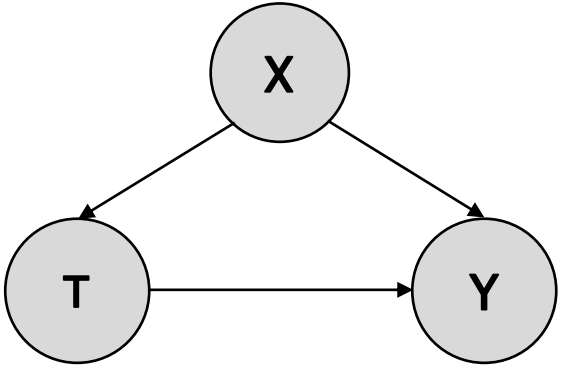
Causal graph and Confounder

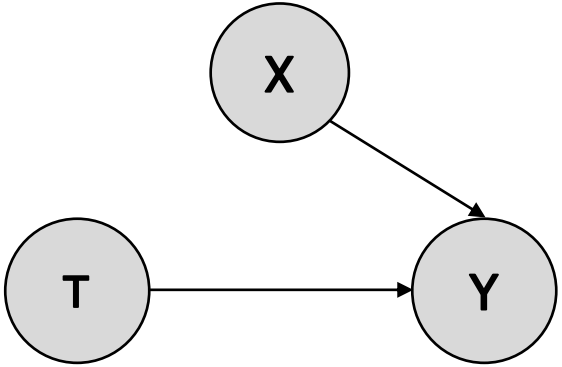


- People only took medicine if they were already feeling sick
- **Confounder:** Variable that affects both probability of receiving treatment and outcome

How can we measure ATE without this problem?

- Randomized control trial (RCT)
- More realistic scenario:
  - We'll probably study effects of medicine on someone who is sick
  - If we survey people, there still might be differences: lower income person may not be able to afford medicine and may also have worse nutrition that leads to more severe illness: income is a confounder (X)
- Instead of surveying people, we take a group of people and randomly assign them to "treatment" or "control" group





Randomized Control Trials

- How can we conduct randomized control trials for:
  - Effects of smoking on lung cancer
    - We randomly assign people to smoke or not smoke?
  - Effect of gender on hiring decisions: do men get more higher paying job offers?
    - We randomly assign people genders??

- Many of the questions we may want to study are not possible to investigate through randomized control trials
- We have to rely on ***observational*** data

References and Acknowledgements

- Anjalie Field, Chan Young Park, Kevin Z. Lin, and Yulia Tsvetkov. "Controlled Analyses of Social Biases in Wikipedia Bios" WebConf (2022)
- Chan Young Park*, Xinru Yan*, Anjalie Field*, and Yulia Tsvetkov. "Multilingual Contextual Affective Analysis of LGBT People Portrayals in Wikipedia" ICWSM (2021)

- Hypothesis testing slides were inspired by slides from Diyi Yang and David Bamman